COMPUTER VISION

Course Assignment

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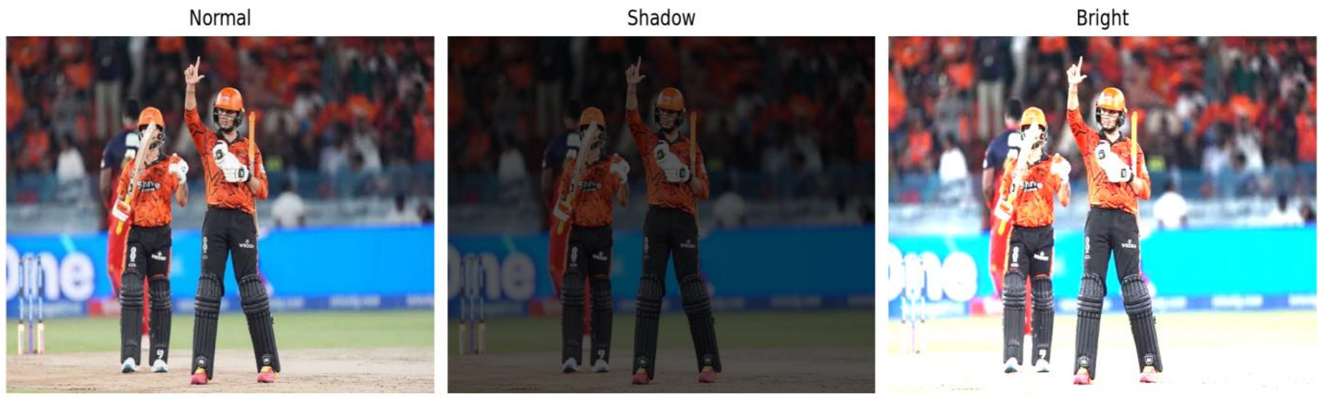
Course Code	ITA0513
Course Name	Computer Vision

See the world. Understand it.
Build the future.

Module 1: Image Representation, Types of Image Processing & Benefits

Exercise 2: Color Space Robustness under Ambient Drift

Task: Convert RGB frames to YCbCr/HSV color space to isolate the luminance component (Y or V) from chromaticity, evaluating tracking stability under variable room lighting.



Result: Isolating the luminance channel left the color channels invariant to lighting drift, successfully preserving tracking across the output sequence of the original image, shadow image, and bright image.

Module 2: Image Preprocessing (Linear vs. Non-Linear Filtering)

Exercise 3: Noise Characterization and Filter Selection

Task: Evaluate a 9×9 Gaussian filter against a 9×9 Median filter on the simulated lighting frames corrupted with random white Gaussian noise and binary salt-and-pepper impulse spikes.

ORIGINAL IMAGE

SHADOW IMAGE

BRIGHT IMAGE

FILTERED IMAGE

Result: The Median filter completely eliminated the salt-and-pepper spikes while preserving sharp geometric object edges, whereas the Gaussian filter failed to remove the impulse noise, blurred the object boundaries, and introduced a distinct halo artifact due to local weighted averaging.

Module 3: Edge Detection (Spatial Gradient Analytics)

Exercise 5: First-Order vs. Second-Order Gradients

Task: Compare 90-degree rotated Sobel (\(G_x, G_y\)) directional gradients against an omnidirectional Laplacian filter on a high-contrast printed target.

ORIGINAL IMAGE

SOBEL X-AXIS

SOBEL Y-AXIS

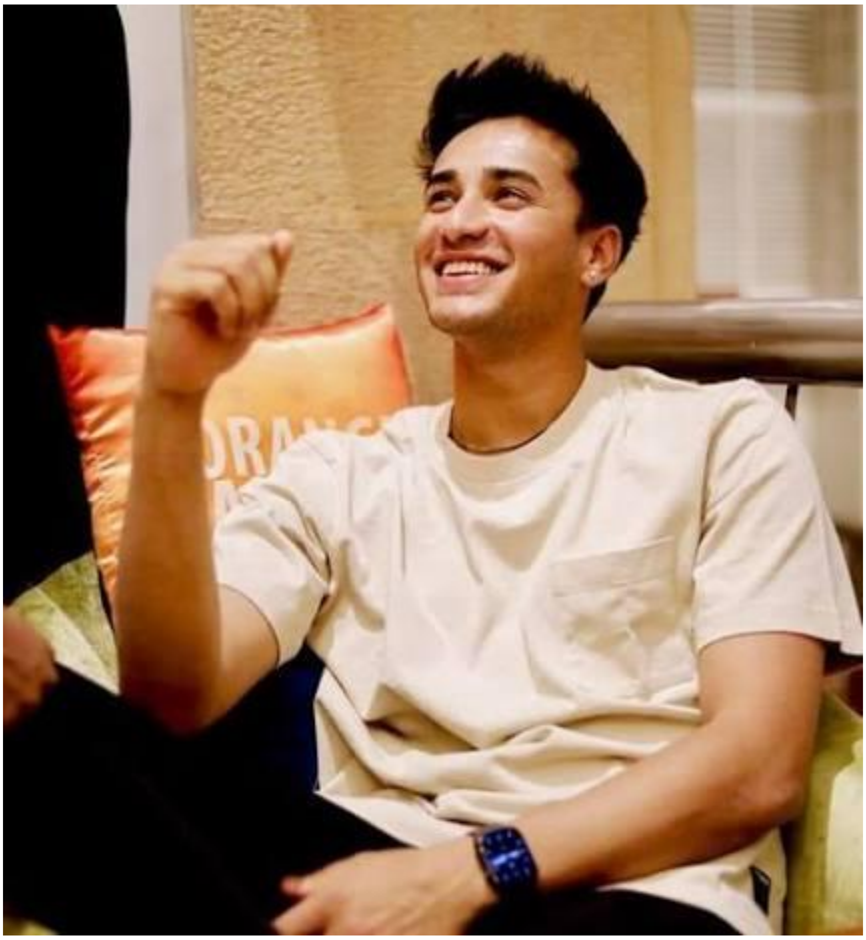
LAPLACIAN IMAGE

Result: Sobel filters isolated specific vertical and horizontal boundaries based on orientation, while the Laplacian successfully mapped all object outlines simultaneously across a single unified on-screen grid.

Module 4: Corner Detection (Harris Spatial Tensor Analytics)

Exercise 7: Eigenvalue Mapping of the Second-Moment Matrix

Task: Evaluate a 2 × 2 Second-Moment Tensor matrix and map principal eigenvalue states (\(\lambda_1, \lambda_2\)) on input_module4.jpg across flat surfaces, straight edges, and intersecting corners.



INPUT IMAGE

OUTPUT IMAGE

Result: Flat surfaces yielded negligible values, straight edges produced a single dominant eigenvalue, and 90-degree intersections generated two balanced, large eigenvalues, allowing the system to plot bright red tracking markers exclusively on valid corners.